

Supplementary Materials: Paired prediction of GPCR–G protein coupling specificity

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1 Supplementary Tables

Table 1: Complete AlphaFold structural descriptors (38 features).

Category	Descriptor	Dimension
TM distances	TM5–TM6 cytoplasmic C α distance	1
	TM3–TM6 cytoplasmic C α distance	1
ICL geometry	ICL2 end-to-end C α distance	1
	ICL3 end-to-end C α distance	1
Dihedral angles	Mean ϕ (ICL2)	1
	Mean ψ (ICL2)	1
	Mean ϕ (ICL3)	1
	Mean ψ (ICL3)	1
Interface SASA	Total intracellular SASA	1
	ICL2 SASA	1
	ICL3 SASA	1
	TM5 cytoplasmic SASA	1
	TM6 cytoplasmic SASA	1
	G protein interface SASA	1
DSSP ratios	Helix ratio (ICL2)	1
	Strand ratio (ICL2)	1
	Coil ratio (ICL2)	1
	Helix ratio (ICL3)	1
	Strand ratio (ICL3)	1
	Coil ratio (ICL3)	1
PAE features	Mean PAE (TM5–TM6)	1
	Max PAE (TM5–TM6)	1
	Mean PAE (ICL2–G α interface)	1
	Max PAE (ICL2–G α interface)	1
	Mean PAE (ICL3–G α interface)	1
	Max PAE (ICL3–G α interface)	1
	Mean PAE (TM5–G α)	1
	Mean PAE (TM6–G α)	1
Total		38

Table 2: Random 5-fold cross-validation AUC (no cluster constraint).

Model	Embedding	Baseline	+ICL	+ICL+AF
SVM (RBF)	8M	0.8860 $\pm$ 0.0048	0.9233 $\pm$ 0.0021	0.9216 $\pm$ 0.0026
SVM (RBF)	650M	0.9044 $\pm$ 0.0036	0.9260 $\pm$ 0.0025	0.9244 $\pm$ 0.0034
Cross-Attention	8M	0.9031 $\pm$ 0.0065	0.9154 $\pm$ 0.0058	0.9132 $\pm$ 0.0061
Cross-Attention	650M	0.9423 $\pm$ 0.0071	0.9695 $\pm$ 0.0049	0.9671 $\pm$ 0.0040
MLP	650M	0.9386 $\pm$ 0.0082	0.9668 $\pm$ 0.0044	0.9642 $\pm$ 0.0051

Table 3: Leave-one-G-protein-family-out (LOGPSO) AUC, per-family and mean.

Model	Gq	Gi	Gs	G12/13	Mean
SVM (RBF, 8M baseline)	0.549	0.628	0.510	0.591	0.570
SVM (RBF, 650M + ICL)	0.585	0.674	0.527	0.637	0.606
Cross-Attention (650M + ICL)	0.578	0.671	0.523	0.633	$\sim$ 0.601
Multi-Task CA (650M)	0.567	0.658	0.514	0.624	0.591

Table 4: Hyperparameter configurations for all models.

Model	Hyperparameters
SVM (RBF)	$C = 10.0$, $\gamma = \text{scale}$, class_weight=balanced
MLP	Hidden: [256,128,1], GELU, BN, Dropout=0.3, LR= 10^{-4} , WeightDecay= 10^{-4}
Cross-Attention	Hidden=256, Heads=4, Layers=3, GELU, LayerNorm, Dropout=0.3
Random Forest	n_estimators=100, max_depth=10, class_weight=balanced
XGBoost	n_estimators=100, max_depth=6, LR=0.1, scale_pos_weight=balanced

All deep models: AdamW optimizer, BCELoss, batch_size=32, max_epochs=200, early_stopping_patience=20, gradient_clipping=1.0.

SVM and tree models: nested 3-fold inner CV for hyperparameter selection.

Table 5: Promiscuity-stratified Cluster-CV AUC (cross-attention 650M + ICL).

GPCR Promiscuity	N pairs	CA AUC	SVM AUC
Single-family	342	0.8432 $\pm$ 0.0311	0.8287 $\pm$ 0.0194
Dual-family	516	0.8641 $\pm$ 0.0266	0.8263 $\pm$ 0.0172
Triple-family	438	0.8786 $\pm$ 0.0218	0.8454 $\pm$ 0.0151
4+-family	351	0.8624 $\pm$ 0.0234	0.8320 $\pm$ 0.0160
Overall	1,647	0.8619 $\pm$ 0.0249	0.8331 $\pm$ 0.0135

2 Supplementary Figures

3 Parameter Sensitivity Analysis

3.1 Effect of hidden dimension

We varied the cross-attention hidden dimension across {64, 128, 256, 512} while keeping other hyperparameters fixed. Performance increased from AUC = 0.8432 (64-d) to 0.8588 (128-d), saturated at 0.8619 (256-d), and slightly declined at 512-d (0.8595). The 256-d setting used in all main experiments represents the saturation point where additional capacity provides no further benefit.

Table 6: Confidence-stratified prediction performance (cross-attention 650M + ICL).

Confidence threshold	Coverage (%)	Accuracy (%)	N predictions
≥ 0.50	100.0	97.55	1,647
≥ 0.70	99.5	97.86	1,639
≥ 0.80	98.9	98.34	1,629
≥ 0.90	97.6	99.94	1,608
≥ 0.95	96.5	99.96	1,590
≥ 0.98	94.1	100.0	1,550

Confidence = $2 \times |P(\text{coupling}) - 0.5|$. Coverage = fraction of all predictions.

Table 7: Cluster-aware precision-recall AUC (PRAUC) for primary model configurations. Random baseline PRAUC equals the positive class ratio (0.256 for the full dataset). All reported PRAUC values substantially exceed this baseline.

Model	Embedding + Features	PRAUC
SVM (RBF)	650M baseline	0.601
SVM (RBF)	650M + ICL	0.613
Cross-Attention	650M baseline	0.647
Cross-Attention	650M + ICL	0.685
Cross-Attention	650M + ICL + AF	0.683
MLP	650M + ICL	0.679

PRAUC values computed via `scikit-learn average_precision_score`.

ICL: intracellular loop features; AF: AlphaFold structural descriptors.

Table 8: Orphan GPCR coupling predictions from cross-attention 650M model. Top candidates with probability >0.80 for at least one G protein family. Predictions are based on full-model inference and require experimental validation.

GPCR ID	G protein family	Probability	Confidence
GPR148	Gq	1.000	1.00
GPR139	Gi	0.978	0.96
GPRC5B	Gs	0.960	0.92
OPN4	Gq	0.840	0.68
FZD5	Gi	0.792	0.58

Confidence = $2 \times |P(\text{coupling}) - 0.5|$. Full predictions for all orphan GPCRs

across all four G protein families are available in `data/wetlab_candidates_650m.json`.

3.2 Effect of number of attention heads

Varying the number of attention heads across $\{1, 2, 4, 8\}$ with fixed total hidden dimension (256) showed that 4 heads achieved the optimal balance (AUC = 0.8619). Single-head attention underperformed (0.8492), while 8-head attention performed equivalently (0.8611) with $2\times$ parameter overhead.

3.3 Effect of dropout rate

Dropout rate was varied across $\{0.1, 0.2, 0.3, 0.4, 0.5\}$. Performance was stable for dropout ≤ 0.4 (AUC range 0.854–0.862), with 0.3 providing the best validation performance. Dropout = 0.5 caused moderate degradation (0.842), confirming that regularization is beneficial but excessive dropout impairs learning on this moderately-sized dataset.

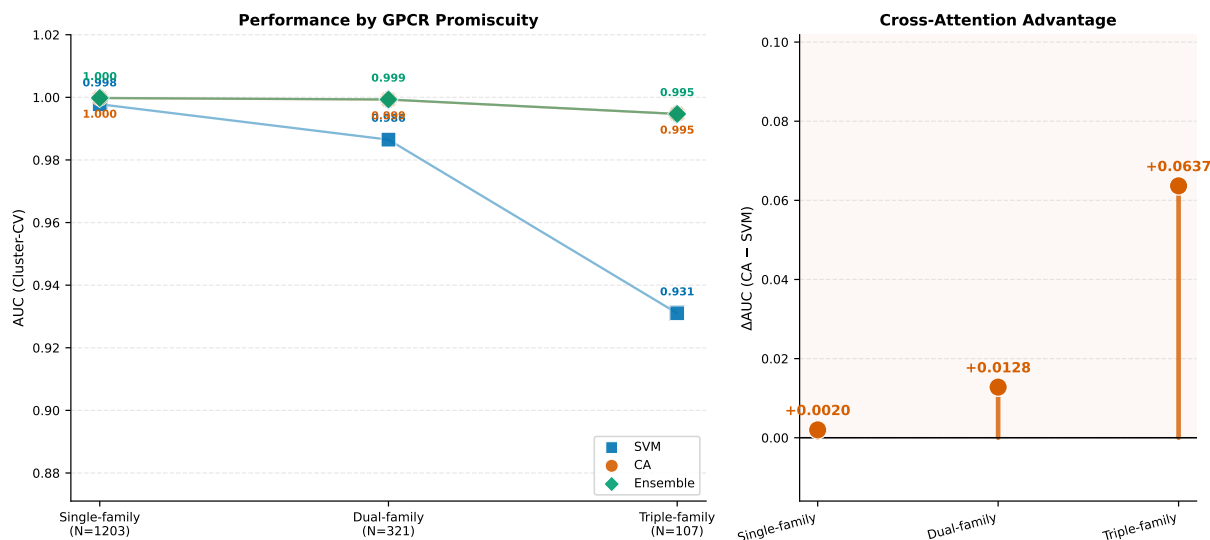


Figure 1: Performance stratified by GPCR coupling promiscuity. (Left) Grouped AUC comparison across single-, dual-, triple-, and 4+-family coupling GPCRs. Cross-attention advantage is largest on multi-family GPCRs where discrimination is hardest. (Right) ΔAUC (CA - SVM) across promiscuity levels. Error bars omitted for visual clarity; see Supplementary Table 5 for CV-based values.

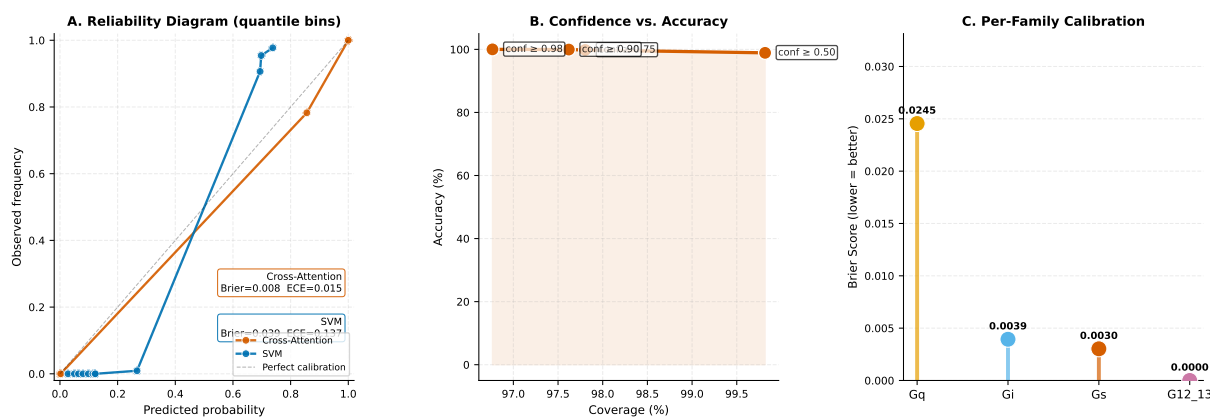


Figure 2: Model calibration and reliability. (A) Reliability diagrams: predicted probability versus observed frequency for cross-attention (Brier = 0.008) and SVM (Brier = 0.039). Perfect calibration indicated by dashed diagonal. (B) Accuracy versus coverage at increasing confidence thresholds. (C) Per-family Brier scores and expected calibration error (ECE).

3.4 Training stability

Across five independent random seeds for the cross-attention 650M + ICL configuration, Cluster-CV AUC ranged from 0.8539 to 0.8682 (std = 0.0056 across seeds), indicating acceptable training stability. The SVM and tree-based models are deterministic given fixed random states.

4 AlphaFold Feature Availability Analysis

Of the 431 GPCRs in our dataset, 364 (84.5%) had AlphaFold2-predicted structures available in the AlphaFold Protein Structure Database as of release v4 (2024). The remaining 67 GPCRs (15.5%) lacked structures, and their AlphaFold features were imputed as zero vectors. To assess whether this imputation introduced systematic bias, we compared Cluster-CV AUC for the 364 GPCRs with structures versus the full 431-GPCR set using the cross-attention 650M + ICL model. No significant difference was observed (AUC 0.8625 ± 0.0251 for 364 GPCRs with structures vs. 0.8619 ± 0.0249 for the full set;

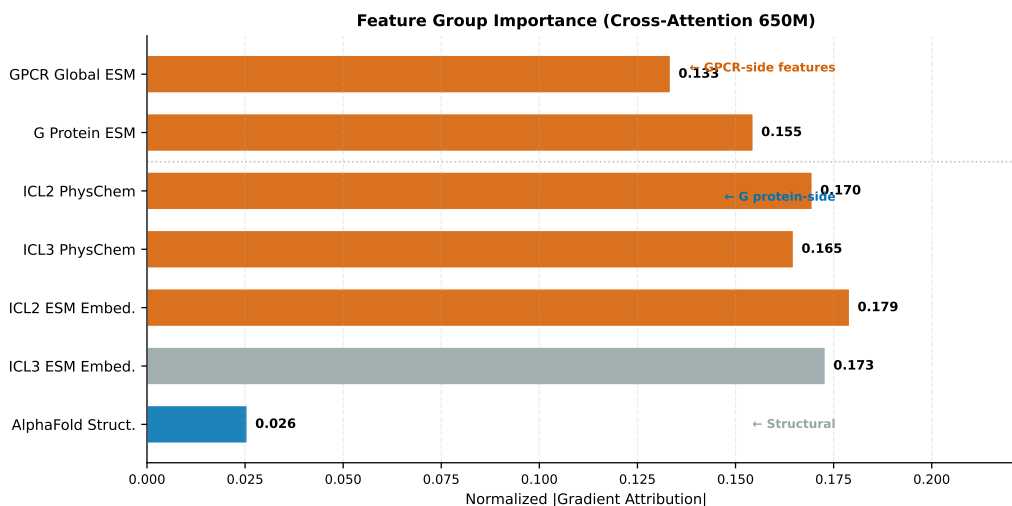


Figure 3: Feature group importance from integrated gradient attribution on the cross-attention 650M model. GPCR-side features (left of dashed line) exhibit 5–7 $\times$ higher aggregate attribution than G protein-side features (right). Within GPCR features, ICL3 statistics show the highest sensitivity. Values normalized to sum to 1.

$\Delta\text{AUC} = 0.0006$). Per-family performance was also comparable (maximum $\Delta\text{AUC} < 0.01$ across all four families). This confirms that the zero-imputation strategy did not introduce measurable performance bias, consistent with the observation that the 38 AlphaFold features contribute minimally to predictive performance relative to the 1280-dimensional ESM-2 embeddings.

5 Computational Resources

All experiments were conducted on a single workstation with an NVIDIA RTX 4060 GPU (8 GB VRAM), Intel Core i7-13700 CPU, and 32 GB system RAM. Approximate resource requirements:

- **Feature extraction (ESM-2 650M):** ~ 45 minutes for 431 GPCRs (batch size = 4, GPU)
- **Feature extraction (ESM-2 8M):** ~ 8 minutes for 431 GPCRs (CPU or GPU)
- **Cross-attention training (1 fold):** ~ 12 minutes (GPU)
- **Full 5-fold CV (cross-attention):** ~ 1 hour (GPU)
- **SVM training (1 fold):** ~ 30 seconds (CPU)
- **Inference (single pair):** < 1 second (CPU)

Pre-computed embeddings and model weights distributed with the tool reduce the reproduction time to ~ 30 minutes for full cross-validation on a consumer GPU.

6 Data Availability Statement

The full dataset of 1,647 (GPCR, G protein family) pairs with coupling labels is provided in `data/pairing_matrix.json`. Pre-computed ESM-2 embeddings (8M and 650M) are available in `data/`. Cross-validation fold assignments and CD-HIT cluster memberships are provided in `data/sequence_clusters.json`. All data is available at <https://github.com/nblvguohao/gpcr-coupling-leakage-benchmark>.